

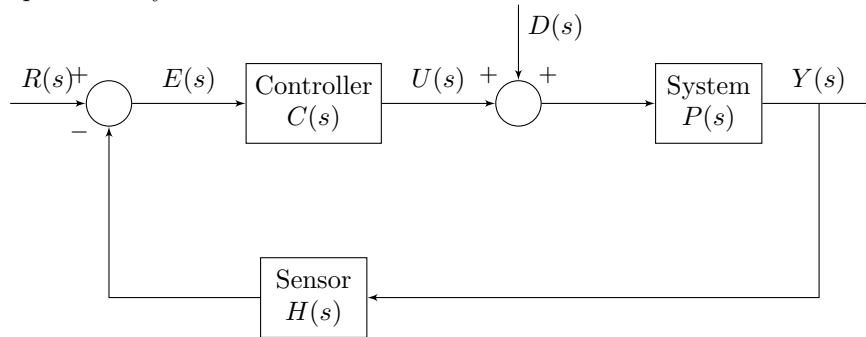
1 January 9, 2026 (Class 2)

System

Disturbance

Feedback loop

open close System



2 January 12, 2026 (Class 3)

2.1 Linear Constant Coefficient Ordinary Differential Equations (LCC-ODE)

A differential equation of the form

$$\sum_{k=0}^n a_k \frac{d^k y(t)}{dt^k} = \sum_{m=0}^M b_m \frac{d^m r(t)}{dt^m},$$

where $y(t)$ is the output (response) of the system and $r(t)$ is the input to the system; and a_i and b_i are constant.

2.2 Laplace Transform

The Laplace transform of a function $f(t)$, defined for all real numbers $t \geq 0$, is the function $F(s)$, which is a unilateral transform defined by

$$\mathcal{L}[f](s) = F(s) = \int_{0^-}^{\infty} f(t)e^{-st}dt,$$

where $s = \sigma + i\omega$ complex frequency-domain parameter, and, 0^- is the shorthand notation for

$$\lim_{\varepsilon \rightarrow 0^+} \int_{-\varepsilon}^{\infty}.$$

2.2.1 Region of Convergence (RoC)

2.2.2 Properties of Laplace Transform

2.2.3 Standard Laplace Transform Pairs

Function	Time Domain $f(t)$	Laplace Domain $F(s)$	ROC
Impulse	$\delta(t)$	1	All s
Step	$u(t)$	$\frac{1}{s}$	$\Re(s) > 0$
Ramp	$tu(t)$	$\frac{1}{s^2}$	$\Re(s) > 0$
Power	$t^n u(t)$	$\frac{n!}{s^{n+1}}$	$\Re(s) > 0$
Exponential	$e^{-at}u(t)$	$\frac{1}{s+a}$	$\Re(s) > -a$
Sine	$\sin(\omega t)u(t)$	$\frac{\omega}{s^2 + \omega^2}$	$\Re(s) > 0$
Cosine	$\cos(\omega t)u(t)$	$\frac{s}{s^2 + \omega^2}$	$\Re(s) > 0$

3 January 16, 2025 (Class 4)

4 January 19, 2025 (Class 5)

5 January 20, 2025 (Class 6)

5.1 Second Order System

We have $G(s)$ as,

$$G(s) = \frac{b}{s^2 + as + b}$$

and

$$\zeta := \frac{a}{2\sqrt{b}} \quad w_n := \sqrt{b}.$$

From this

$$G(s) = \frac{w_n^2}{s^2 + 2\zeta w_n s + w_n^2},$$

with poles as

$$\sigma_{1,2} = -\zeta w_n \pm w_n \sqrt{\zeta^2 - 1}.$$

5.1.1 Relation between ζ and the step response

For $\zeta > 1$: Poles are real.